restricted assumption is that we want to establish a baseline for the further development of such processing procedures and we also want to identify how far the existing natural language processing and general-purpose programming techniques can achieve the goal.

Our approach is inspired by an observation that is intrinsically related to the notion of the degree of interest (DOI). According to the commonly known explanation of DOI, variations of perceived details in a scene are the function of the viewer's interest. The function reflects where the viewer's interest is placed. Usually, we pay more attention to things next to us. In contrast, we may pay less and less attention to things further and further away from us. If we turn this thinking to natural languages, we recognize something strikingly similar – *variations of descriptive details in text are the function of the writer's interest*. If we are writing about a few topics, we tend to use naturally more words to describe, clarify, differentiate, and iterate topics that we think are more important than the rest of them. We tend to find more examples and take into account more perspectives than otherwise. As a result, the more important topics will be surrounded by far richer varieties of words than the rest of topics. The design of the new procedure draws upon this observation and focuses on identifying the core of such a concentration in text as the symbol of a concept. In addition, we decide to concentrate on the predicate chain of the subject, the verb, and the object in a sentence so as to simplify the original sentence and make it easy for the user or the analyst to decide whether there is sufficient interest to pursue. We expect that given a sufficiently large amount of input, it becomes more likely that the basic structure of a sentence is to be found more and more frequently, especially for important topics. As such instances accumulate, emergent patterns may appear. Such patterns are expected to be insightful for making sense of unstructured text. They may play instrumental roles in facilitating the further visual analysis of seemingly unstructured text.

Procedure

The flow chart in Figure 7.11 illustrates the key components of the procedure and their communications.

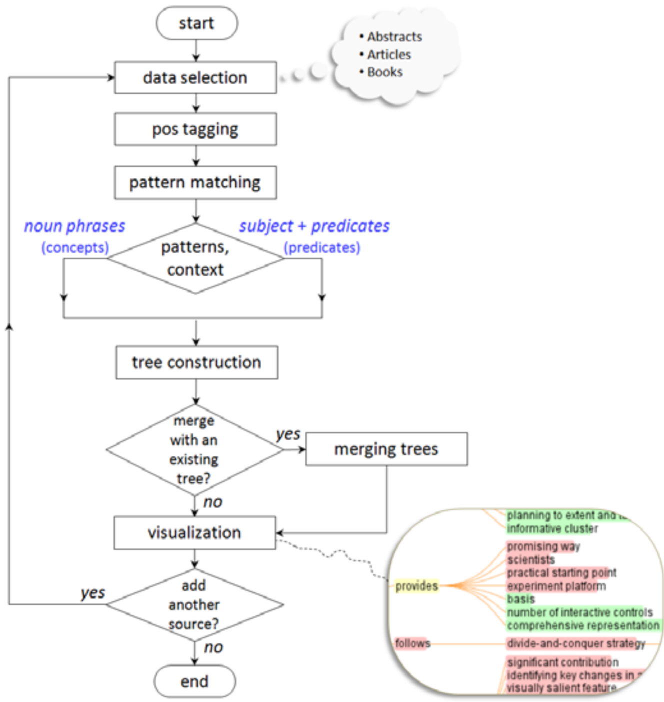

Figure 7.11. The flow chart of the procedure of the method.

The procedure starts with the selection of sources of text. The user may select a single document, multiple documents, and a directory of documents as the initial input data source. The selected documents as a while are referred as a source of the procedure. The current prototype supports two sources. The text in the selected source is subsequently processed by part-of-speech (POS) tagging. The result is that each and every word in the original text is annotated with a POS tag. For example, the noun tree is tagged as tree/nn and the verb run is tagged as grow/vb.

The next step, pattern matching, is to identify segments of word sequences based on their POS tags and then organize various parts of such segments according to heuristics on implied hierarchical relations. For example, the noun phrase *large-scale network* can be split into two parts *large-scale* and *network*. The noun *network* is known as the head noun. The *large-scale* part is known as the modifier of the head noun. Thus the word network represents a concept and it will be stored as a parent node in a hierarchical representation. The term *large-scale* can be seen as an attribute of the concept and it will be stored as the child node to the parent network. The pattern matching process is illustrated in Figure 7.12.

Table 7.6 summarizes the major patterns defined by regular expressions over POS-tagged text. To make the construction of these patterns easier, we use a bottom-up approach. Starting with the basic building blocks such as nouns, verbs, and adjectives, more complex patterns are built by joining these building blocks. For example, the predicate pattern is defined in terms of subject, verb, and object, which are in turn defined in terms of noun phrases and verb groups.

The length of a regular expression pattern can serve as an interesting benchmark. A future improvement of the method may shorten the length while maintaining the overall coverage and accuracy of the extraction and pattern matching step. Furthermore, more complex and lengthy patterns may be designed to capture more subtle and complex assertions in unstructured text.

A concept tree and a predicate tree will be generated as the results of the pattern matching step. The concept tree consists of all extended noun phrases found in the text. Phrases with the same head noun are aligned up first, then their attribute words. For example, *heterogeneous information space* and *exploration of an information space* share the concept of information space. Both *heterogeneous* and *exploration* are organized as the children nodes of information space as its attributes. Similarly, if a predicate of *we* + *introduce* + *a new algorithm* is found, the construction algorithm will first check if there is already a node we in the predicate tree. If the we node exists, then check if it has a child node named *introduce*, then finally check the next level for a possible place for the term *new algorithm*. Leading a's, an's, and the's are omitted from the tree representations.

Different trees of the same type can be merged by the same rules. One may also incorporate more advanced graph mining techniques to find identical substructures in such trees. Furthermore, a predicate tree can be refined and enhanced by normalizing its subjective nouns and objective nouns with concept nodes in a concept tree. For example, the predicate *we* + *introduce* + *a new algorithm* can be standardized as *we* + *introduce* + *algorithm*.

The merge step integrates tree representations generated from multiple sources but retains the identification of each of the original sources so that one can compare the contributions from different sources side by side. Merging patterns from different sources follows the same rules for within-source constructions, except that the origin source of each pattern is now taken into account. Internally, patterns from the first source and those from the second source can be distinguished. Three colors are used to encode the three possible relationships: pink - two patterns are from the same first source, green - both are from the same second source, and yellow - they have different sources. In principle, one may choose to allow the addition of a series of sources, although it may considerably increase the cognitive burden of the analyst to recognize patterns from too many possible sources.

The final step is for users to interact with visualizations and explore the underlying data through the hierarchical representation of concepts and predicates. Using tree representations can leverage many existing tree visualization tools. For example, prefuse supports a number of ways to visualize a tree structure, including the balloon layout, the radial layout, and the node-and-link layout. We experimented different layouts and finally decided that the node-and-link layout provides the most appropriate design option for us.

The prefuse implementation of the DOITree provides a robust vehicle for the baseline prototype. It allows the user to zoom in and out easily. It scales reasonably well. The response rates decrease for concept trees with more than 200,000 nodes. On the other hand, pruning trees may become advisable because it is likely to increase the clarity and the speed of interaction.

The analyst is able to access the original context of each instance associated with a node by moving the mouse cursor over the node. The contextual information shows the names of the sources and the complete sentences in which a pattern is found.

Table 7.6. Building blocks of patterns.

Pattern	Definition	Example	Length(Pattern)
noun	(article OR adjective)* + word /nn[sp]*	heavy and cold rain	181 char
noun phrase	various combinations of nouns and other types of words	Information visualization research	858 char
subject	noun OR noun phrase OR word/prp	this article	1,057 char
simple verb	word /vb[dzpn[^g]]	discover	27 char
verb	Various combinations of verbs	could have been discovered	257 char
concept	noun phrase, including noun of noun	exotic plant	858 char
predicate	subject + verb + (noun phrase OR noun OR word/vbg)	we + introduce + a new algorithm	3,480 char

Figure 7.12. The regular expression of the subject-predicate pattern consists of 3,480 characters. The sentence on the top of the figure is tagged first. Corresponding patterns are matched based on rules defined in the regular expression for predicates. Identified patterns are added to the tree.

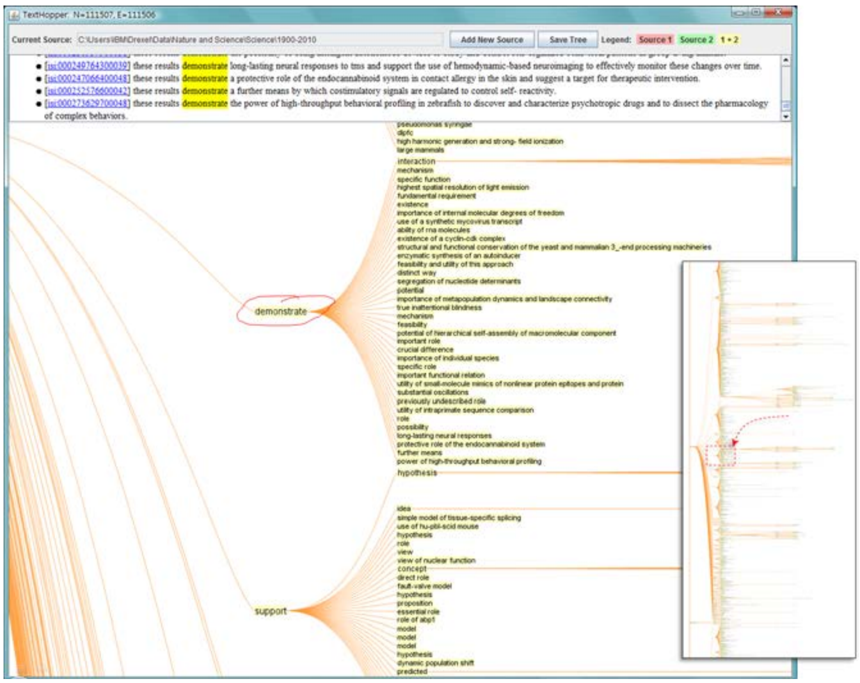

Figure 7.13. The user interface of a prototype. The portion of the predicate tree shown in the figure represents the pattern of <u>these results demonstrate …</u>, which forms a small branch of a 111,507-node hierarchy constructed based on 110-year *Science* article abstracts (1900-2010).

The user interface of a prototype is shown in Figure 7.13. It shows a predicate tree that was constructed from 110-year *Science* articles' abstracts (1900-2010). The predicate tree consists of 111,507 nodes, including both nouns and verbs that form a predicate. The circled term *demonstrate* is part of the predicate pattern: *these results demonstrate*. The full sentences in context are displayed on the top of the screen and instances of the current focal node in the tree are highlighted in yellow.

Examples of Use

We tested the prototype on three types of text, namely abstracts of scientific papers, abstracts of patents, full-length articles, and monographs. In particular, abstracts of scientific papers include IEEE InfoVis papers (2000-2009), *Science* (1900-2010). Abstracts of patents include 823 *Google* patents and 15,227 *Yahoo*! Patterns. Full-length articles include Shneiderman's highly cited *The Eyes Have It* (Shneiderman, 1996), and two journal papers by Chen on *CiteSpace* (Chen, 2004; Chen, 2006). The prototype was applied to two books: one is the *Brokerage and Closure* by Burt (Burt, 2005) and the other is Charles Darwin's 1872 edition of *The Origin of Species* (6^{th} ed.) (Darwin, 1872), excluding the glossaries and index of the book. We intend to use this set of examples to set a baseline for subsequent evaluations.

We recorded the total number of sentences and words found in each text, the percentage of nouns and verbs identified by part-of-speech tagging, and the sizes of both the concept tree and the predicate tree. We are particularly interested in the relationship between the average length of sentences and the overall coverage rate (the percentage of sentences that are found with concept and predicate patterns). We also recorded the runtime taken to completion. The experiment used an IBM ThinkPad T500 with duo processors of 2.53GHz and 3GB of RAM and the Java Runtime version of 1.6.0_11. The results are shown in Tables 7.7 and 7.8.

Table 7.7 Datasets tested.

Source	Type	Sentences	Words	Nouns (%)	Verbs (%)
Yahoo Patents(15227)	Abstract	9,342	189,662	40.72	13.84
Google Patents (823)	Abstract	4,372	83,586	39.40	14.40
InfoVis (2000-2009)	Abstract	2,139	42,472	34.40	12.35
Darwin (1872)*	Book	5635	197,332	23.05	13.77
Burt (2005)	Book	5,201	112,556	30.94	12.87
Science (1900-2000)	Abstract	98,370	2,062,010	37.09	10.98
Chen(2004)	Article	358	6440	30.45	13.23
Shneiderman(1996)	Article	258	4,500	34.04	12.51
Chen (2006)	Article	659	10,831	33.60	12.55

*Excluding glossaries and the index.

Table 7.8 Records are sorted by the coverage rate (% sentences in trees).

Source	Type	Concepts	Predicates	Coverage (% sentences)	Runtime (mill. Sec)
Yahoo Patents(15227)	Abstract	15,227	12,082	67.90	71,666
Google Patents (823)	Abstract	7,091	5,393	63.63	577
InfoVis (2000-2009)	Abstract	5,364	2,535	58.85	6,957
Darwin (1872)*	Book	13,583	5,538	50.22	279,622
Burt (2005)	Book	11,187	5,896	49.51	147,826
Science (1900-2000)	Abstract	279,932	111,506	49.44	4,852
Chen(2004)	Article	899	375	41.06	8,722
Shneiderman(1996)	Article	747	279	37.60	6,514
Chen (2006)	Article	1,463	589	34.90	14,756

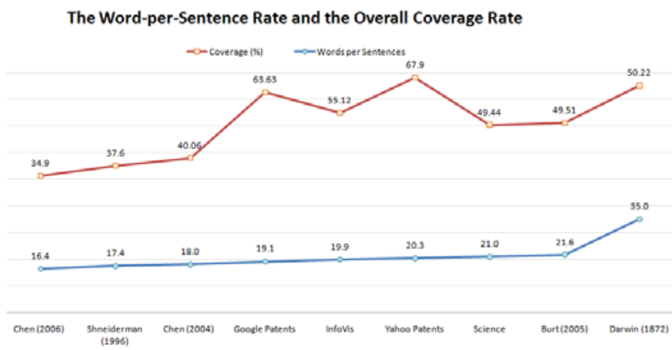

Figure 7.14. Words from longer sentences may be more likely to be included in the trees.

Figure 7.14 suggests that the average length of sentences in text may be correlated with the overall coverage rate. Patent abstracts and InfoVis abstracts have particularly higher coverage rates than other sample datasets tested. It seems to suggest a general trend of the higher word-per-sentence rate, the higher the coverage with the three exceptions in the middle. A possible explanation is that all the ones in the middle are primarily information science and computer science related. In contrast, the *Science* abstract dataset is more in line with others perhaps because of its multidisciplinary nature and that the performance of the pos-tagger may be deteriorated for biomedical literature. Yahoo and Google patent abstracts yielded the highest coverage rate, followed InfoVis abstracts. The two books are placed in the middle of the coverage range. Individual full-length articles have the lowest coverage rate among all the sample datasets.

Scenarios of Use

Three major scenarios of use are outlined as follows, although one can certainly use the method in many ways.

The first scenario is the analysis of a multi-document collection. The analyst needs to identify the major topics and claims along with their original contexts. The analyst may have a few options. At a macroscopic level, she may choose to visualize inter-document relations based on similarity measures derived from term frequencies and decompose the entire collection into topically related clusters of documents. At a microscopic level, she may also use tools such as Phrase Nets and Word Tree to develop an understanding of the content at word and sentence levels. Between the two levels, she may explore the concept tree and the predicate tree to obtain the information that would enable her to make better use of the tools operating at the other two levels. For example, as shown in Figure 7.15, the major topics of InfoVis are shown as the nodes with many children nodes in its concept tree, such as *data*, *information*, and *visualization*. Furthermore, the analyst can zoom in and out to obtain more contextual details of a concept, for example, all sorts of adjectives and nouns that were found surrounding the term *data* in this particular source of text.

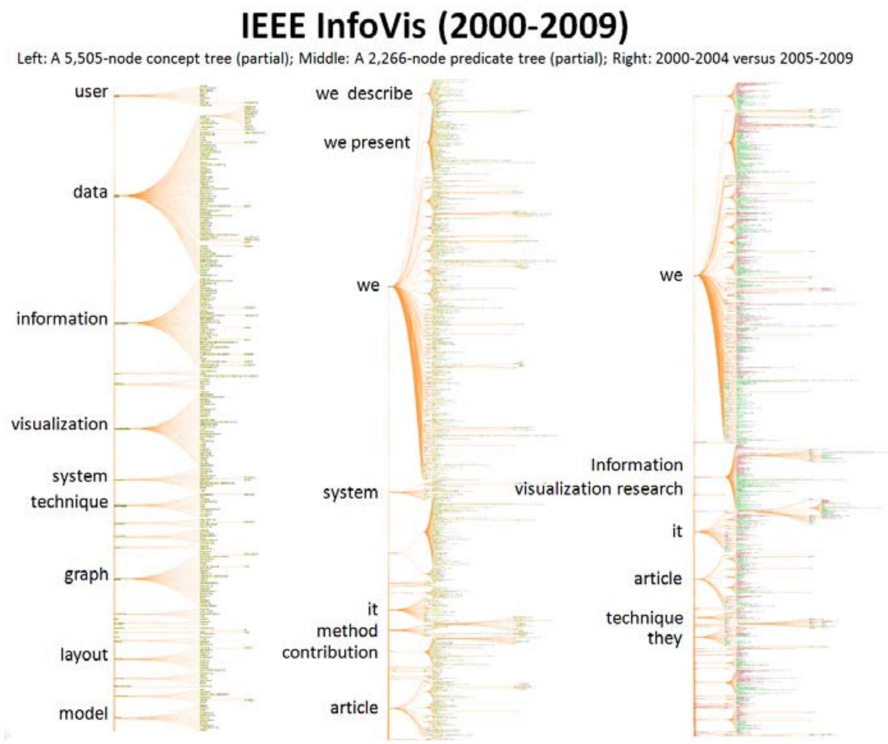

Figure 7.15. A concept tree, a predicate tree, and a comparative predicate tree of the InfoVis abstract dataset.

One potential advantage of using a tree representation over alternatives such as a network is the improved clarity and readability. Major topics are visually prominent in tree representations without risking potential problems of distractive cross links that are often seen in network representations of such datasets. Another potentially significant advantage is that the analyst may quickly track down the most dominating rhetorical patterns in the data source. Combining with the knowledge of what major topics are, the analyst can browse through the entire collection without being restricted to the boundaries of individual documents; in fact, the grouping patterns allow the analyst to compare similar assertions across different documents in the collection. In the current prototype, one can find all the instances of a pattern by moving the mouse mover the node in the tree representation.

The second scenario is the comparison of two sources or two samples from the same source. If we want to know how InfoVis papers in the earlier years differ from InfoVis papers in the later years, we can split the 10-year dataset into two parts, generate the trees for the first part, and add the secondary part to these trees. Figure 7.16 is an example of such merged trees. Patterns found in the first sample are shown in pink; those found in the second sample shown in green; and patterns in common are in yellow. These predicates are associated with the subject node *article*. We can see that the most commonly used patterns in the InfoVis dataset include *article* + *presents* + *, *article* + *introduces* + *, *article* + *proposes* + *, and *article* + *describes* + *. These rhetorical statements are probably also common for the majority scholarly publications. Scrolling deeper down the tree reveals more semantically focused statements, for example, *GPU* + *requires* + *data parallel programming*.

In addition to collection-collection comparisons, one may need to compare two documents, or one document against one collection. The notion of source makes it flexible. One can instantiate a source with one document or multiple documents. This flexibility allows an analyst to apply standard clustering algorithms on a collection of text and then apply the new approach introduced here to each cluster and compare different structures by treating clusters as incoming text sources. The new approach provides an alternative to make sense individual clusters by using techniques such as text summarization. The advantage of the visual approach over traditional text summarization is that the analyst can move back and forth smoothly across different levels of words, sentences, documents, and the entire data set, which would reduce the cognitive burden of the analyst.

The third scenario is the study of a lengthy document. In this scenario, the analyst can use the interactive visualization of the concepts and predicates as an indexing mechanism. Since all the instances and contexts of a concept can be found within the subtree of the concept, it becomes easy for the analyst to access them. We illustrate this scenario with two book examples.

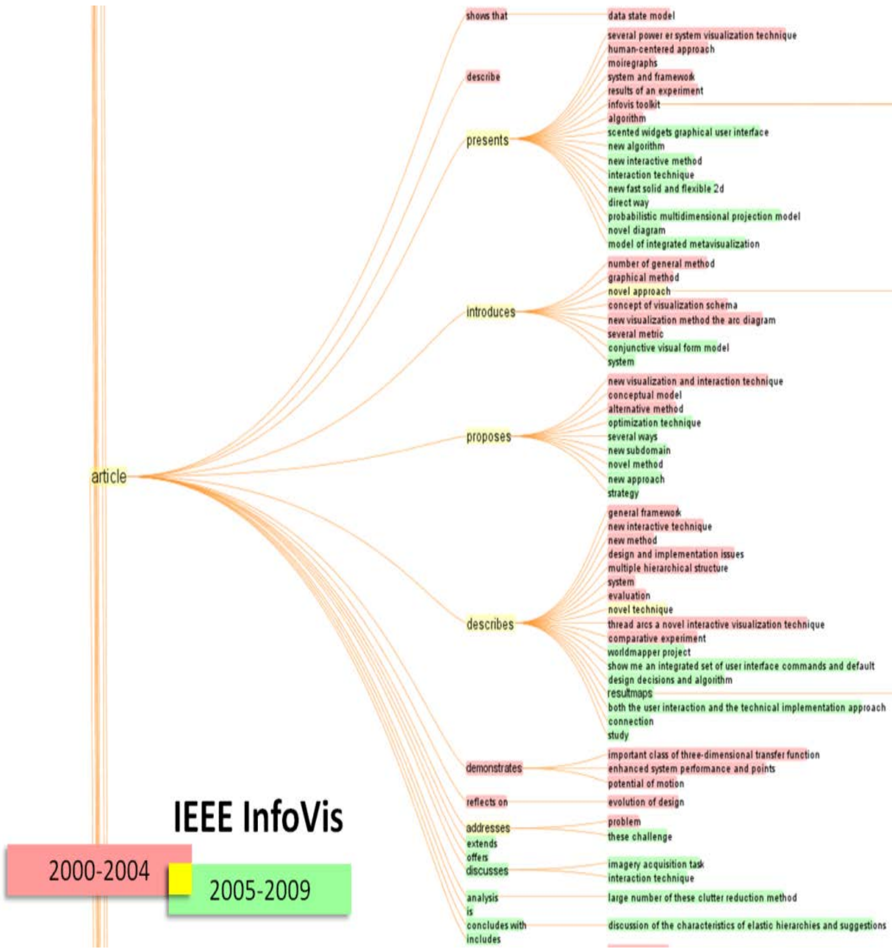

Figure 7.16. A merged predicate tree from IEEE InfoVis papers in two periods of time: 2000-2004 and 2005-2009.

Brokerage and Closure is written by sociologist Ronald S. Burt (2005) to introduce the concepts of brokerage and closure in social networks and their practical implications. Figure 7.17 shows the top of the concept tree and the top of the predicate tree, where prominent patterns are usually positioned. For example, it is visually evident that the book has many ways to describe *people*, *network*, *trust*, and *ideas* as shown on the left concept tree. Similarly, the predicate tree on the right reveals that the leading actors in the book are *you*, *we*, *they*, and *it*.

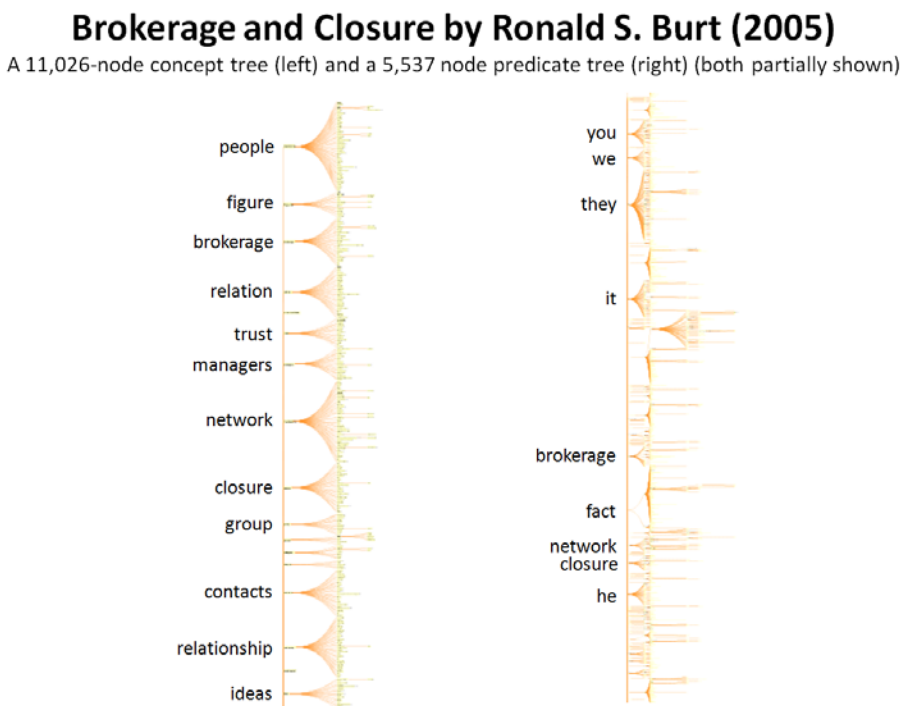

Figure 7.17. A concept tree and a predicate tree of Ronald Burt's 2005 book.

The example in Figure 7.18 is Darwin's classic *The Origin of Species* (6th ed.). Predictably, *species* is the most prominent concept, followed by *forms*, *varieties*, *differences*, *animals*, *plants*, and *group*. Its predicate tree includes patterns such as *forms of life* + *are* + *, *it* + * + *, *we* + * + *, *they* + * + *, and *many exotic plants* + *have* + *.

Further Improvement

Design and technical issues are discussed as follows, which could serve as the basis for further improvement.

The highest coverage rate among the 9 datasets is the Yahoo patent abstracts (67.90%). The lowest coverage rate is a full-length article published in the Journal of the American Society for Information Science and Technology (JASIST) (34.90%). The InfoVis abstract dataset has the 3rd highest coverage (58.85%). The current set of heuristics and regular expression patterns obviously cannot process all the sentences. Some of the problems are due to the POS-tagging errors. Some are due to the complexity of sentence structures.

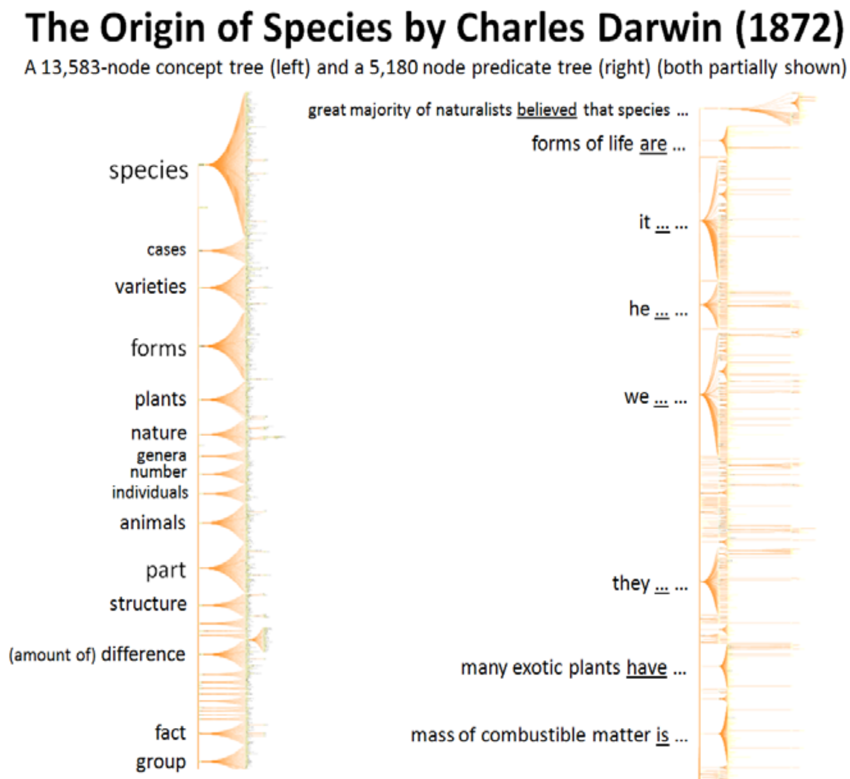

Figure 7.18. A concept tree and a predicate tree of Darwin's *the Origin of Species*.

We list some examples below and hope they can serve as test data for further enriched and enhanced heuristics and pattern matching rules. The challenges for the further improvement include improving the efficiency of the pattern matching mechanisms by shortening the lengths of regular expressions, expanding the coverage of the processing procedure so that a wider variety of sentences can be handled, and improving the time efficiency of the algorithm by shortening the overall runtime.

1. Our/PRP$ tool/NN replaces/VBZ manual/JJ and/CC in/IN some/DT cases/NNS pen-and-paper/JJ based/VBN analysis/NN tasks,/NN and/CC we/PRP discuss/VBP how/WRB user/NN feedback/NN was/VBD incorporated/VBN into/IN iterative/JJ design/NN refinements/NNS
2. The/DT method/NN is/VBZ widely/RB used/VBN in/IN many/JJ research/NN fields/NNS including/VBG biology,/JJ geography,/NN statistics,/NN and/CC data/NN mining/NN
3. However,/NNP most/RBS dendrograms/NNS do/VBP not/RB scale/VB up/RP well,/JJ particularly/RB with/IN respect/NN to/TO problems/NNS of/IN graphical/JJ and/CC cognitive/JJ information/NN overload/NN

4. The/DT overview/NN displays/VBZ only/RB a/DT user-controlled,/JJ limited/JJ number/NN of/IN nodes/NNS that/WDT represent/VBP the/DT "skeleton"/NN of/IN a/DT hierarchy/NN
5. The/DT contribution/NN of/IN the/DT paper/NN includes/VBZ a/DT new/JJ metric/JJ to/TO measure/VB the/DT "importance"/NN of/IN nodes/NNS in/IN a/DT dendrogram;/NN the/DT method/NN to/TO construct/VB the/DT concise/NN overview/NN dendrogram/NN from/IN the/DT dynamically-identified,/JJ important/JJ nodes;/NN and/CC measure/NN for/IN evaluating/VBG the/DT data/NNS abstraction/NN quality/NN for/IN dendrograms/NNS
6. We/PRP evaluate/VBP and/CC compare/VBP the/DT proposed/VBN method/NN to/TO some/DT related/JJ existing/VBG methods,/NN and/CC demonstrating/VBG how/WRB the/DT proposed/VBN method/NN can/MD help/VB users/NNS find/VB interesting/JJ patterns/NNS through/IN a/DT case/NN study/NN on/IN county-level/JJ U.S/NNP

There are more fundamental questions to be addressed. For example, what is the nature of these extracted patterns? Are they more rhetorical than semantic? The top-level predicate patterns appear to be rhetorical such as *we* + *propose* + *a new algorithm* … Since we are interested in facts and hypotheses, we are looking for predicates such as *smoking* + *causes* + *lung cancer*. We found such statements in the 111,506-node predicate tree of Science, but we do not have the statistics of how many of them and do not yet have reliable features that would allow us to distinguish scientific statements from rhetorical ones. Purifying predicate trees becomes necessary in order to build analytical reasoning functions. Another way to refine the structure of predicate trees is to utilize the structure of concept trees, as we mentioned briefly earlier.

The scalability of the new method is also an important issue for further research. The current prototype can process full-length books. It handles the 110-year *Science* abstract dataset surprisingly well, although interacting with the 111,506-node predicate tree and the 279,932-node concept tree experienced considerable delays. The burden on the visualization program is partially due to the built-in storage of contextual information for each node. Using an indexing mechanism to store the bulk of data outside the tree structures would improve the performance.

The implementation of the prototype uses our own hand-crafted regular expression patterns. More sophisticated patterns may be developed and tuned up with natural language processing tools such as GATE. It will be also useful to compare the coverage and other benchmark scores with a wider range of natural language processing resources, including various pos-taggers.

The new method has several application implications. For example, the tree construction procedure can be used to develop an alternative indexing and ranking algorithm by weighing on the size of subtrees of a node. One may also derive semantic metrics based on the positions of nodes on such trees and measure the degree of interest between two nodes. This indexing potential has the advantage of preserving the original context and providing an easy to access interface to all the instances in multiple documents.

The new method provides an extra layer of interface between text visualization and text so that one can focus on exploring aggregated patterns as the intermediate linkage between specific words and sentences and their broader context. This extra layer enables the analyst to move back and forth across the boundaries of documents and focus on the essence of the text.

The emergent structural patterns enable users to identify the areas to pursue from the global overview of the entire dataset. The new method allows the analyst to contrast and compare two sources of text at various levels of detail, for example, in the study of patents from competing corporations or publications from different schools of thoughts.

Future work should address the issues discussed above. In addition, constructing an authoritative ontology of the information visualization field for comprehensive experimental tests, incorporating ontology construction techniques, and conducting user evaluation and field studies are among the promising routes to proceed.

Detecting Abrupt Changes

Michael Jackson, the legendary king of the pop, died on June 25, 2009. The news of his death created a surge of search volume on the Internet that was too much for Google News to handle.[3] The volcanic peak was marked as B in Figure 7.19.

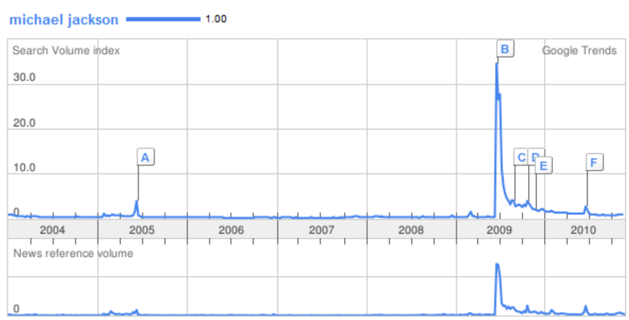

Figure 7.19. The volcanic peak of search of "Michael Jackson." Source: Google Trends.[4]

This type of surge is also known as a burst. Identifying such bursts in a timely manner is the primary goal for *burst detection*. While it is reasonably straightforward to track down reasons behind this type of burst as well as their timing and duration, it can be much more complex and challenging in other situations. Is there a visible burst of positive or negative reviews of *The Da Vinci Code* in Figure 7.1? Was there a burst of particular terms in those reviews? In the literature of a scientific field, do we expect to

[3] http://articles.cnn.com/2009-06-26/tech/michael.jackson.internet_1_google-trends-search-results-michael-jackson?_s=PM:TECH

[4] http://www.google.com/trends?q=michael+jackson

see a burst of a topic as it becomes suddenly popular? If a field is experiencing a Kuhnian paradigm shift, or a conceptual revolution, would we be able to detect a burst of articles representing the new paradigm?

A Burst of Citations

We have addressed some of these questions in earlier chapters in the context of structural analysis and the theory of discovery. In this chapter, we will show how burst detection can be used in comparative studies so that we would be able to answer questions such as whether positive reviews tend to burst before negative reviews, or when and how long the burst of a new paradigm would last.

First, let us look at an example of citation burst. Figure 7.20 shows a part of a larger network of papers that were cited together in the literature of complex network analysis. This part is in fact the so-called largest connected component of the entire network. The node highlighted in the middle was Wasserman's representative work on social network analysis. The node is in a crucial position that connects two sub-networks of the component. It is a pivotal point in the development of the topic. The cluster above the pivotal node is labeled by the term *scale-free network*, the cluster below the pivotal node is labeled by the term *social network*. These two clusters corresponding to the earlier social network analysis that was mostly done by sociologists (the one below) and the more recent complex network analysis that was largely done by physicists.

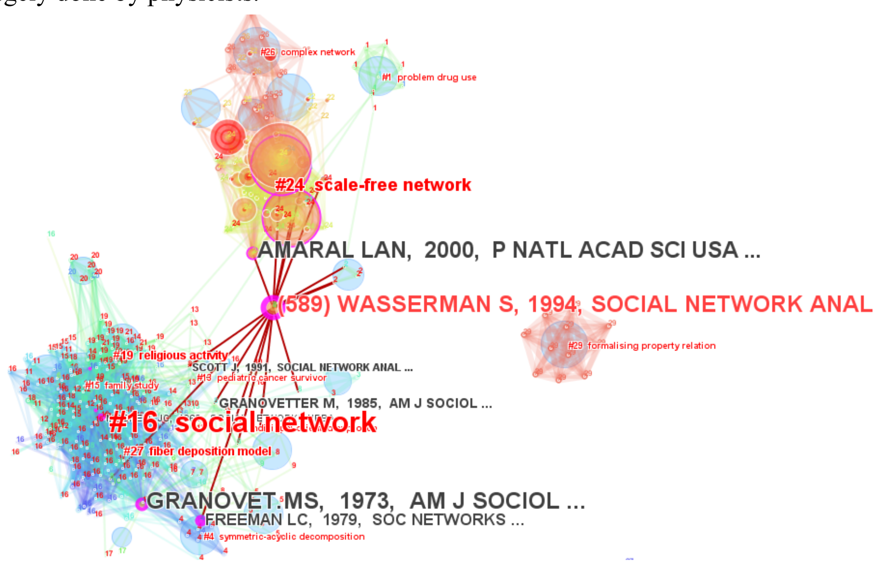

Figure 7.20. The largest connected component of complex network analysis research (1980-2009).

The unique position of the pivotal node suggests that it is essentially one of the few that both communities had in common. Was there any burst of citation to the pivotal-point work? If we can detect a burst, what would it tell us about the evolution of the two communities?

As it turns out, there was indeed a burst of citation. The burst started in 1998 and lasted till 2000 (Figure 7.21). Among other early citers in 1998, one of them was the groundbreaking paper for complex network analysis written by Watts and published in *Nature*. What is intriguing is the position of the burst in the entire 20-year history. The level of the citation counts during the burst period is not the highest on the hindsight, but the timing is much more meaningful for the purpose of identifying an emerging trend or a new field. The timing synchronized with the publication of the groundbreaking paper of the new complex network analysis. Furthermore, it was indeed cited by the groundbreaking paper. The citations of the pivotal work increased rapidly from 2002. The delay in part reflects how long it may take for knowledge diffusion and for the new paradigm to establish.

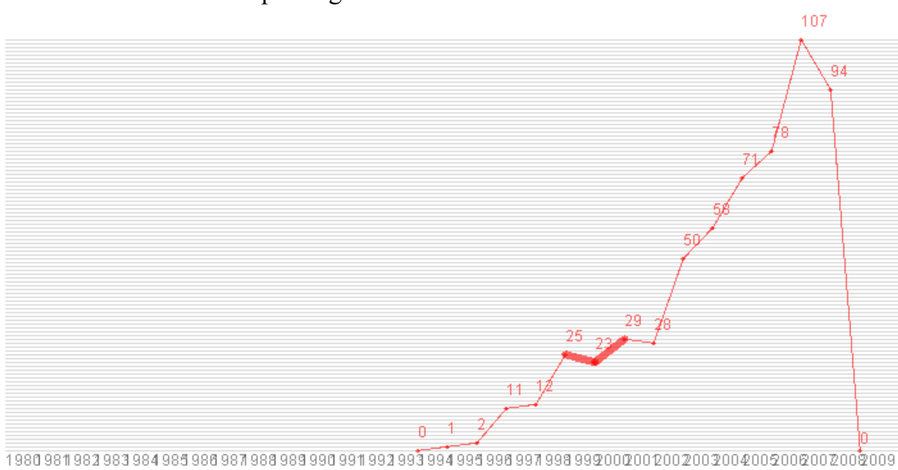

Figure 7.21. A burst of citation was detected between 1998 and 2000.

Survival Analysis of Bursts

Survival analysis is a statistic method for answering questions concerning the time elapsed till the occurrence of an event, which is called *survival time*. Survival analysis looks at the probability of survival at various time points. Survival analysis also takes into account *censored cases*, which refer to cases that never experience the occurrence of an event throughout the observation time or drop-off the observation. A typical use of survival analysis is to compare the effectiveness of a new drug in comparison to that of an existing drug. For example, survival analysis can be used to assess whether a new painkiller is more effective than an existing painkiller. The survival time of a painkiller can be defined as the time elapsed from the time that the medicine is taken till the pain gets back. In this case, the returning pain is the event in question. If patients taking the new drug experience a higher survival probability than the existing drug, then the new drug is more effective.

We are dealing with scientific publications, citation patterns, grant proposals, and other types of knowledge representations. We are mainly concerned with a burst of citations to a published paper and a burst of the frequency of a topic. We can analyze the probability of survival at any time in terms of two types of events associated with a burst: 1) time elapsed prior to burst and 2) the length of the duration of a burst. Figure 7.22 illustrates the two types of comparisons. Time to effect is the waiting time prior to the initial burst. Thus, the shorter the waiting time, the better it is. The same principle should hold for both citation bursts and topic bursts. Similarly, a longer duration of a citation burst is more preferable. So is a longer duration of a topical burst.

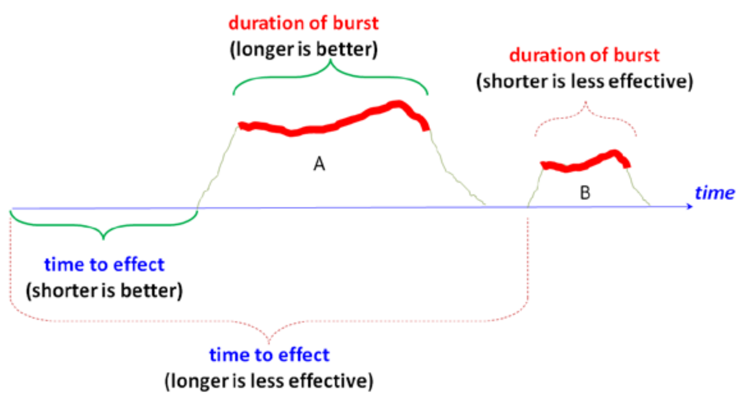

Figure 7.22. Comparing time to effect and duration of burst with survival analysis.

Figure 7.23 illustrates a concrete example of the citation history of a highly cited paper in string theory written by Juan Maldacena. The paper was published in 1998. A citation burst was detected between 1999 and 2003. The waiting time was 1 year. A total of 288 cited references are shown in the network in Figure 7.24. They are split into two groups by the median citation of 19.50, high- and low-citation groups, so that survival analysis can address the question whether papers the two groups differ in terms of patterns associated with their citation bursts.

Figure 7.23. The waiting time and the duration of a citation burst for a 1998 paper by Maldacena. The burst began in 1999, one year after its publication, and ended in 2003.

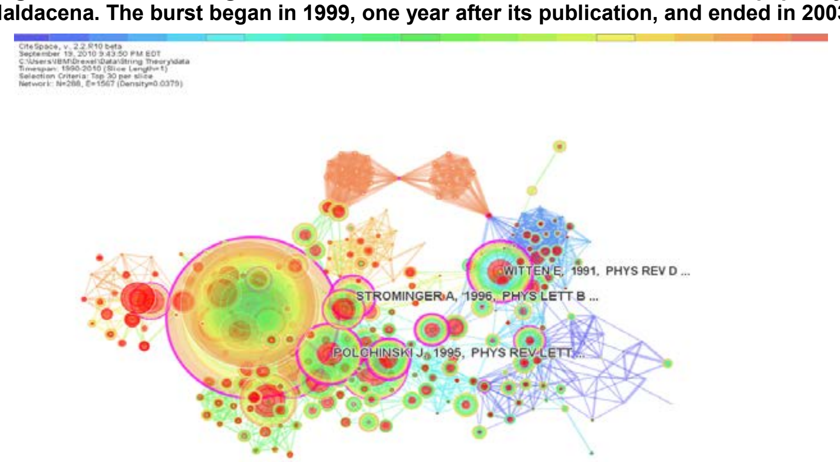

Figure 7.24. The input data for survival analysis consists of 288 cited references as shown in this network. They are split into a high- and low-citation group by h-index.

Survival analysis found that the highly cited reference group is likely to have a citation burst sooner than the less cited papers. A highly cited paper on the average waits for about 2.2 years, whereas a less cited paper waits for 5.2 years (See Figure 7.25). A highly cited paper tends to experience a burst of citation sooner than a less cited paper.

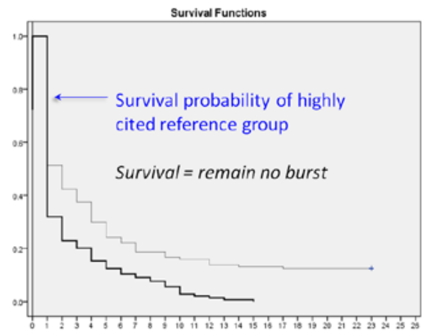

Figure 7.25. The highly cited papers are likely to have a citation burst sooner (burst within 2.2 years) than the less cited papers (burst within 5.2 years).

Differentiating Awarded and Declined Proposals

The previous example suggests that survival analysis, combined with burst detection, can help us to differentiate two kinds of publications, namely papers that are highly cited and papers that are not. If we look closely at the procedure, it appears that the procedure itself is quite generic. The procedure can be used to compare not only two but several groups. A broad range of events can be defined accordingly. Thus, the procedure is applicable to compare different types of proposals, different types of patent applications, as well as different types of scientific publications. In the following example we illustrate how this procedure can be used to differentiate successful and unsuccessful proposals.

We hypothesize that that awarded and declined proposals may differ in terms of how and when they deal hot topics. The timing of the appearances of a hot topic can be measured by burst detection. The comparison between the awarded and declined groups is done by survival analysis, which allows us to address the question whether the two groups differ statistically in terms of how soon hot topics appear and how long they last. The results indicate that this is indeed detectable with statistically significance with the caveat that the choice made in the noun phrase extraction and therefore the text segmentation appears to influence the sensitivity of the analysis.

We considered hypotheses that may distinguish awarded and declined proposals:

1. Awarded proposals address topics in a timelier manner than declined proposals.
2. Awarded proposals address more profound topics than declined proposals.
3. Noun phrases extracted from core segments are more specific and focused on proposed research questions than terms from one-page project summaries.
4. Survival analysis of noun phrases extracted from one-page project summaries is the same as results from noun phrases extracted from the core segments.

The first hypothesis can be tested in terms of the survival probabilities of bursts of terms as the events. Awarded proposals are expected to deal with hot topics earlier than declined ones. The second hypothesis means that the duration of a term burst in an awarded proposal is expected to be longer than the corresponding duration in a declined proposal.

The hypotheses were tested with 1,206 awarded and 4,305 declined proposals from a program of the NSF over four years (2007-2010). The results, shown in Table 7.9, include the number of noun phrases extracted and the number of noun phrases with detected bursts. The "Time till burst" column shows the p-level of the statistical significance between the awarded and declined groups. Notably, awarded and declined groups are distinguishable by single words and two kinds of noun phrases, namely, single-word nouns and noun phrases consisting of up to 4 words. No statistical differences were found using noun phrases with two or more words. The results suggest that awarded and declined proposals differ in their use of single words or short noun phrases.

Table 7.9. Time till bursts of terms in project descriptions of a sample of NSF proposals.

Noun Phrase		**Awarded**			**Declined**			**Time till burst**
min	max	subtotal	bursted	(%)	subtotal	bursted	(%)	p level
1	1	9238	1241	13.4	26358	3928	15	0.001
2	2	6018	591	9.8	21961	2159	9.8	0.990
3	3	4204	338	8%	15809	1295	8	0.760
4	4	4092	323	7.8	15492	1274	8	0.500
1	4	9541	1039	10.9	29394	3721	12.7	0.000
2	4	5743	506	8.8	21064	1939	9	0.367
single word		3963	1059	26.7	7689	3389	44	0.000

Awarded and declined proposals statistically differ in terms of the survival time till the bursts of noun phrases of 1-4 words. Now we further compared whether the one-page project summaries provided by the principal investigators (PIs) and core passages extracted from 15-page project descriptors would be statistically different in terms of survival time till bursts of noun phrases of 1-4 words. More detailed descriptions of how we extracted core passages will be given in Chapter 8.

We extracted core passages from 5,412 proposals (1,150 awarded and 4,262 declined). The average length of the top-1 core segments (762 words) is longer than their corresponding project summaries (613 words). The difference is statistically significant with a two-way t test, p = 0.000. This justified our assumption that the core information is longer than the 1-page summary but shorter than the 15-page description. In addition, one-page summaries tend to use broader terms, whereas the top-ranked core segments tend to include more specific terms. We found a statistically significant difference between the survival probabilities of awarded and declined proposals (Table 7.10). This means that core segments are better sources of text than 1-page summaries for studies of documents such as grant proposals.

Table 7.10. Awarded and declined proposals differ in terms of the survival time till term bursts in top-ranked core segments.

Source	Noun Phrase	Awarded			Declined			Time till Burst
	min/max	subtotal	bursted	(%)	subtotal	bursted	(%)	p level
1-page summary	1/4	6815	571	8.3	23620	1990	8.4	0.916
top-1 core segment	1/4	9541	1039	10.9	29394	3721	12.7	0.000
summary/segment		71%	55%		80%	54%		

Survival analysis of bursts of noun phrases (1-4 words) from one-page project summaries revealed a statistically significant difference (p=0.007) between awarded and declined proposals in terms of the duration of a burst. As shown in Figure 7.26, bursts of terms in awarded proposals lasted shorter (on average 1.792 year) than in declined proposals (on average 2.381 year), although no statistically significant difference was found.

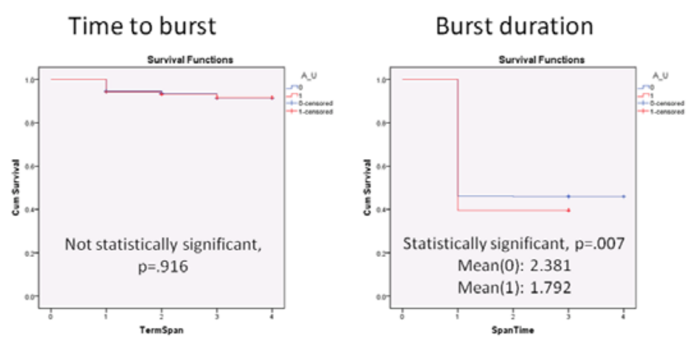

Figure 7.26. Awarded and declined proposals have different survival probabilities of burst duration. Source: one-page project summaries of proposals (1 – awarded, 0 – declined).

Summary

In this chapter, we have first addressed some of the issues concerning how to differentiate conflicting opinions in an evidence-based approach, in particular, the role of decision trees in representing terms that may predict the orientation of customer reviews. The method particularly takes the advantage of the available ratings of reviews. Decision trees of terms and positions of reviewers provide not only a descriptive model of the central issues of a debate but also a predictive model to anticipate the paths of arguments one may follow.

The second part of the chapter deals with situations in which we do not have judgments from users or experts in numerical forms and we do not have a taxonomy or ontology either. We have introduced a method we are developing to address these issues by tapping on patterns that we can discern from linguistic relations. The flexibility needed to tolerate the ambiguity of natural language becomes available when concepts and predicates are organized in hierarchical structures.

The first part of the chapter is concerned with the role of temporal patterns such as bursts in differentiating multiple groups of documents in different categories. Survival analysis of the timing and duration of citations and term use in scientific publications and grant proposals has demonstrated the potential of this method.

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